Huawei NetEngine AR6700V-L Virtual Router Data Sheet

The NetEngine AR6700V-L is a new-generation virtual router developed by Huawei for the cloud era. It can be flexibly deployed in virtual and public cloud environments and can be used as a virtual CPE or virtual gateway in Huawei SD-WAN solutions. It can also be used as a cloud access gateway to meet various network requirements of enterprises in cloud scenarios.

Product Overview

The benefits brought by cloud, virtualization, and NFV technologies are well known. The AR6700V-L is a software AR router deployed in virtualization and cloud environments as a VNF. It is based on the industry-leading Huawei network software platform that is the same as the AR hardware router. Compared with the previous-generation products, the provides more network and security functions, stronger forwarding performance, and more flexible deployment capabilities, meeting enterprises' network requirements in SD-WAN, SASE, and cloud scenarios.

The AR6700V-L can be deployed on the Hypervisor of the Point of Presence (PoP), enterprise headquarters, enterprise branches (including offline and cloud branches), and public cloud environments, providing flexible and quick deployment options for enterprises. Expand the enterprise WAN to the cloud. With the Huawei SD-WAN solution, unified security and management policies can be implemented with the physical network, achieving unified and better SD-WAN network experience.

Key Features and Benefits

Professional and rich SD-WAN network and security capabilities

- Shares the industry-leading professional network software platform VRP with AR hardware routers, providing rich network functions and consistent O&M experience.
- Supports multiple tunnel types, such as EVPN and SRv6. Functions as the SD-WAN IWG to interconnect with traditional MPLS networks and supports SD-WAN multi-gateway interconnection, meeting various complex networking requirements.
- Seamlessly interconnects with public cloud network services and supports cloud network scenarios such as host VPC, transit VPC, and cloud router, providing better cloud migration experience.
- Secure SD-WAN, IPsec, IKEv2, and AES-CBC/AES-GCM/SM encryption algorithms are supported to ensure network security.

Flexible architecture and powerful performance

- The control plane is separated from the data plane. Multi-core CPUs are supported. Computing resources can be deployed on demand based on service loads, reducing resource costs.

- The data plane is optimized for virtualized and cloud-based environments, providing stronger forwarding performance for encrypted traffic. Unique High-Pass technology supports higher single-stream bandwidth, better meeting AI computing network requirements.

Flexible and automated deployment

- Supports server- and uCPE-based Hypervisor deployment and mainstream public cloud deployment.
- Cloud site deployment for one-click automatic deployment.

Huawei iMaster-NCE supports unified management, automatic deployment, and visualized O&M.

Figure 1 shows the system architecture of the NetEngine AR6700V-L.

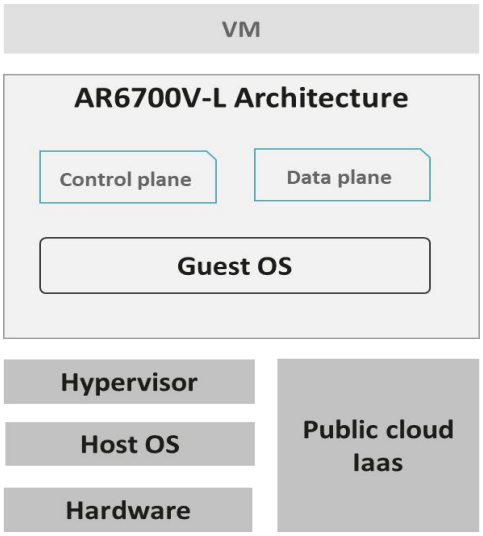


Figure 1 NetEngine AR6700V-L system architecture

Product Specifications

Feature Specifications

Name	Description
Hypervisor deployment	Linux KVM, VMWare ESXi, OpenStack, ZStack
Public cloud deployment	HUAWEI Cloud, AWS, Alibaba Cloud, Azure, Tencent Cloud, Oracle Cloud, Tianyi Cloud, Volcano Engine, Google Cloud, China Mobile Cloud
IO model	SR-IOV, VirtIO, PCI-Passthrough
VM Specifications	CPU：2/4/8/16vCPU RAM：4/8/16GB Storage：38G
Routers	Static route, OSPFv2/v3, BGP/BGP4+, routing policy, policy-based routing

Address management	ARP/ND, DHCPv4/DHCPv6, DNSv4/DNSv6, NAT
VPN	L3VPN, SD-WAN EVPN, GRE VPN
WAN optimization	A-FEC, Data compression
Multicast	IGMP V1/V2/V3, PIM SM, PIM DM, MBGP, IP multicast over SD-WAN
Security	ACL/ACLv6, AAA, RADIUS, IPsec, IKEv2, AES-CBC/AES-GCM/SM Encryption Algorithm, FireWall,IPS*, URL filtering*, Antivirus*
QoS	Traffic policing (CAR), traffic shaping, MQC (traffic classification, traffic behavior, and traffic policy), three-level scheduling and three-level shaping (Hierarchical QoS), priority mapping, congestion management, and congestion avoidance, Intelligent Traffic Policing
Reliability	VRRP, HRP, BFD
Management and O&M	Device management, NETCONF/Yang, CLI, NTP, NQA, Telemetry, and NetStream

*: * Only the AR6700V-L with 4vCPU/8vCPU/16vCPU supports this function.

Performance data

Specifications	AR6700V-L (2vCPU)	AR6700V-L (4vCPU)	AR6700V-L (8vCPU)	AR6700V-L (16vCPU)
RAM	4 GB	8 GB	8 GB	16 GB
SD-WAN IPsec performance* (1400byte)	700 Mbps	7 Gbps	14 Gbps	29 Gbps
SD-WAN typical performance** (IMIX)	-	2 Gbps	4 Gbps	7 Gbps
SD-WAN typical performance** (512byte)	-	2.3 Gbps	4 Gbps	9 Gbps
SD-WAN typical performance** (1400byte)	-	5.8 Gbps	10.9 Gbps	22.9 Gbps

Test environment for the preceding performance data: RH2288X V5 server; CPU: Intel(R) Xeon(R) Gold 6238R CPU @ 2.20 GHz; Network adapter: Intel X710, SR-IOV.

*: The SD-WAN IPsec performance model is EVPN + IPsec(AES-256-CBC), peak performance. This specification applies to the combined total of traffic in inbound and outbound directions.

**: Scenario where the typical SD-WAN performance specification applies: IPsec + QoS + SA + AppFlow(AES-256-CBC), peak performance. This specification applies to the combined total of traffic in inbound and outbound directions. The IMIX forwarding performance in Info-Finder is removed from R25C10 and is now displayed as the forwarding performance of packets with a size of 512 bytes. If you need the forwarding performance information of packets with a size of 512 bytes, you can switch to a version earlier than R25C00.

Typical Application

vCPE in the SD-WAN solution (on-cloud or offline branch)

Enterprise cloudification has become a common trend in enterprise digital transformation. Both on-cloud branches and offline branches are important components of the IT infrastructure of multi-branch enterprises. Huawei's SD-WAN solution can connect multiple branches of an enterprise to build a stable and efficient enterprise WAN with powerful functions, unified management, consistent experience, and high efficiency.

The AR6700V-L can be conveniently deployed on the IaaS of the public cloud and private cloud in branches on the cloud. As the vCPE of the SD-WAN, the AR6700V-L can be connected to the VPC on the cloud in host VPC, transit VPC, or connected to the router on the cloud, enables branches on the cloud to connect to the SD-WAN network for unified management and consistent network experience with offline branches. The AR6700V-L can be flexibly deployed on offline branch servers or uCPEs to function as vCPEs to connect offline branches to the enterprise SD-WAN network.

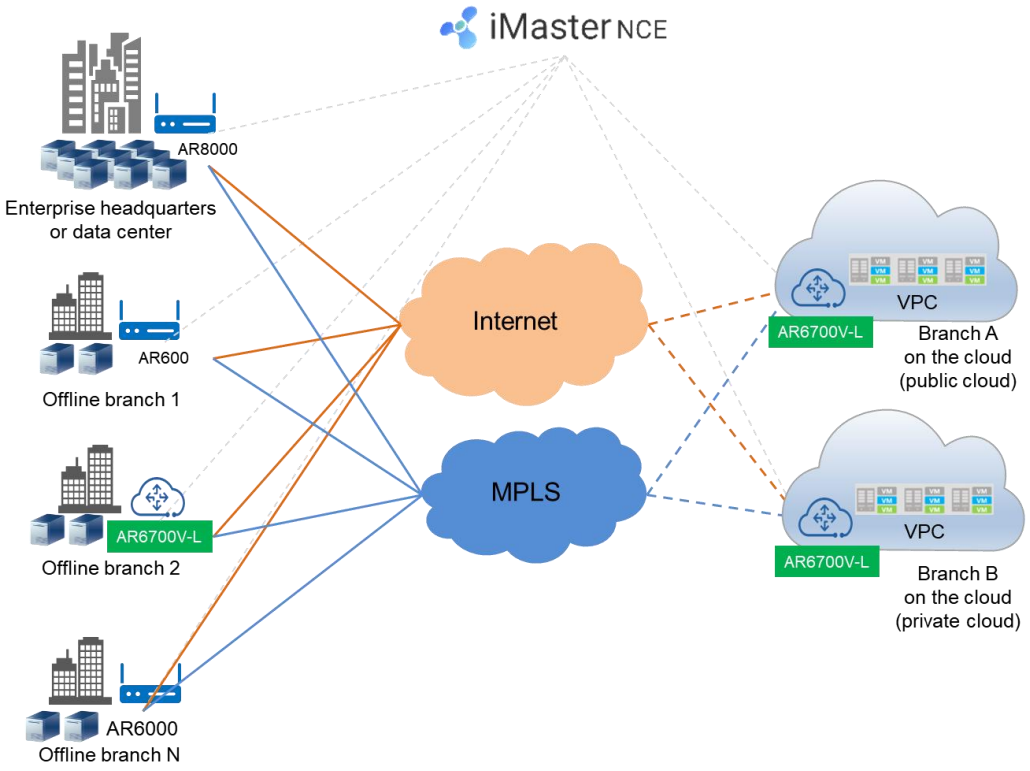


Figure 2 The AR6700V-L functions as the vCPE of the cloud branch and offline branch.

More information

For more information, please visit the Huawei website. http://enterprise.huawei.com/cn/or theContact us by:

Global service hotline: http://e.huawei.com/en/service-hotline Huawei technical support:

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